



Geo localisation Web Science

COMPSCI4077/COMPSCI5107/COMPSCI5078

Joemon M Jose

Overall info

announcements

Communication

- Through Teams Page
 - so please sign in -
 - [COMPSCI4077, COMPSCI5078 & COMPSCI5107: WEB SCIENCE - 2023-24 | General | Microsoft Teams](#)
 - If you are not on Teams Channel
 - you will not get any information/emails
- Labs & Coursework
 - Distributed in Teams File area
 - **Select** File – class materials – coursework or labs
- **Lecture materials**
 - Moodle
 - <https://moodle.gla.ac.uk/course/view.php?id=44995>

Communication

- Through Teams Page
 - so please sign in -
 - [COMPSCI4077, COMPSCI5078 & COMPSCI5107: WEB SCIENCE - 2023-24 | General | Microsoft Teams](#)
 - If you are not on Teams Channel
 - you will not get any information/emails
- Lecture materials
 - Moodle
 - <https://moodle.gla.ac.uk/course/view.php?id=44995>
- Coursework & deadlines
 - Announced in Teams File Area – Files/class materials/courseworks/
 - Read them – geolocalisation lecture third week – Reddit Lecture on week 4 – ask questions during the lecture

Which module to enroll?

- COMPSCI4077 - Web Science H – For level H students
 - Level H students are allowed to do limited M level modules. If you choose to do Web Science MSc module, please enrol for it
- COMPSCI5107 - Web Science for MSc
- COMPSCI5078 - Web Science M – For Level M students
- Please enrol for the correct module
 - That fit with your course plan

Please watch the Introduction lecture

- If you have any questions, please write your questions at
 - https://glamys.sharepoint.com/personal/joemon_jose_glasgow_ac_uk/_layouts/15/doc.aspx?sourcedoc={9185079d-37bf-48dd-84db-aa34e6aeb910}&action=edit
 - I will answer them in the class
 - This will help rest of the class as well
 - they will get an opportunity to clarify concepts

Exams

- Exams contribute 80% to the final grade
- Follow the rules of School of Computing Science
 - Which is based on the University of Glasgow Guidelines
- Online and Supervised
 - Open book
- Hence, solving similar but open problems
 - Need to have a better understanding of the topics so that
 - You should be able to apply solutions intelligently
- Two exams
 - May and August
 - If you fail in May, take Resit exam in August
 - Resit exams are a lot harder
 - Make sense to avoid it

Coursework

- Coursework contributes 20% to final grade
- Level M students to do **Reddit Data Analysis**
 - Based on Lecture 5 (3rd February 2025)
 - Msi (Level 5) and MSc
 - **Under no circumstances MSc/Msi students do H-level coursework**
 - Deadline ~~Friday, 7th March 2025 430PM~~
 - **New Deadline** Friday, 28th February 2025 430PM
- Level H students to do **Geo-localisation**
 - Based on Lectures up to geo-localisation (on 27th February 2025)
 - Deadline Friday, 28th February 2025 430PM



Learning Objectives

- What is geo-localisation?
- Why geo-localisation is important?
 - What kind of geo-localisation allocation possible?
- Geo-localisation methods and limitations
- Understand **evaluation methodology**
- **Look at potential bias**

Level H coursework is based on this topic
Announced in Teams- File Area – Class Materials- coursework - Hle'

Understanding location – a generic problem

- Search engines
- Apps
- Focus on textual clues



NEWS ON SCOTLAND @NewsOnScotland · Sep 12, 2017



Car **accident** closes busy **west end** road in both directions
glasgowlive.co.uk/news/glasgow-n... **#glasgow** **#scotland**



Geo-localisation

- Assigning geo-coordinates to non-geotagged tweets
- geo-coordinates?

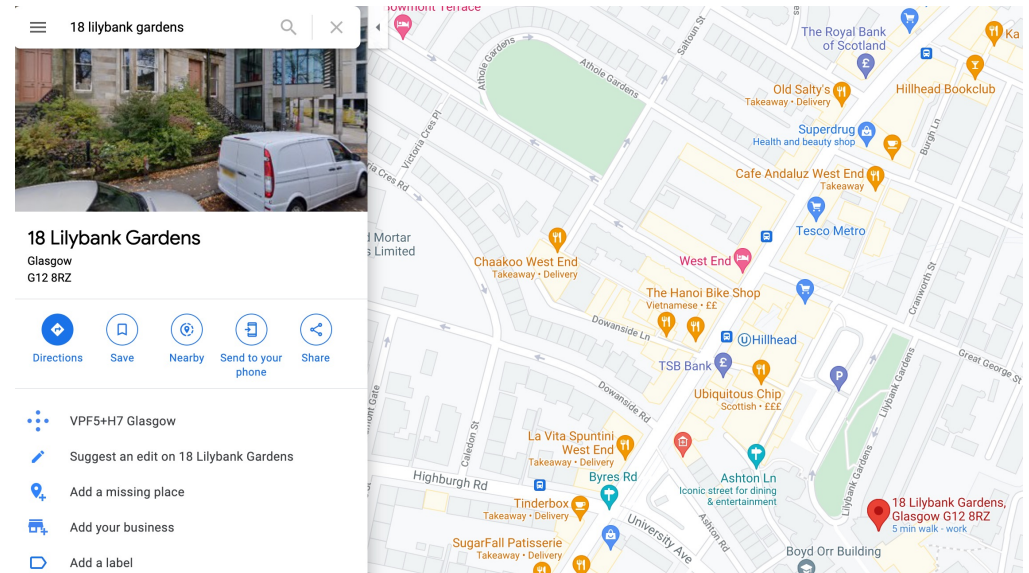
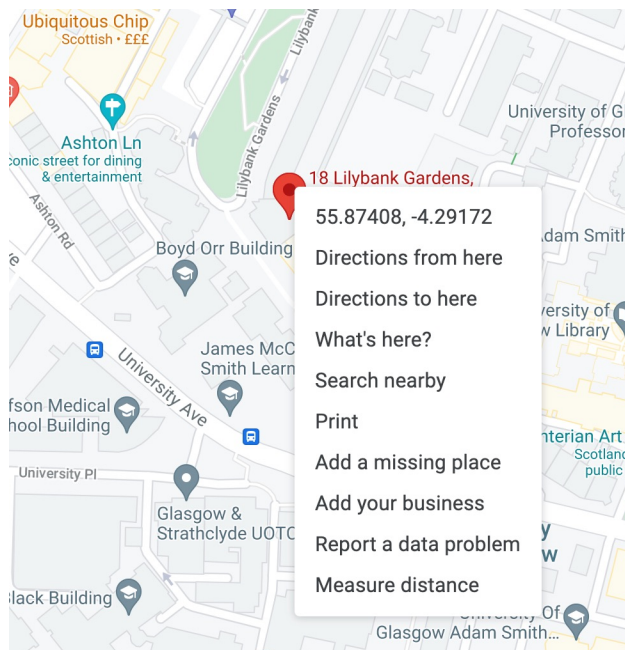
Benefits

- Identifying the location will lead to
 - Quick responses from civil or security services
 - Making alerts to citizens
- Aim is to identify the location as close as possible
 - Byres road vs west-end
 - Fine-grained nature
- Alternative is to use sensors like
 - Traffic cameras
 - Traffic flow measurements

How to find coordinates on Google Maps on the mobile app?

- Open the Google Maps app on your iPhone or Android phone.
- Enter the location or select and hold to drop a pin on the map of the location you want the coordinates for.
- Scroll down to find the coordinates.
- Tap the coordinates to copy to your phone's clipboard.

your latitude coordinate is between -90 and 90.
your longitude coordinate is between -180 and 180.



Geo-location of a tweet

- Profile of the user
 - User location can be extracted from the location field of the user profile
- If geo_enabled = true
 - Coordinates = {long, Lat}



```
in_reply_to_screen_name": null,  
"user": {  
  "id": 3131372879,  
  "id_str": "3131372879",  
  "name": "Uncle Memester",  
  "screen_name": "UncleMemester",  
  "location": "Matthew 24:13",  
  "url": null,  
  "description": "Parody! Fluent in #GenX. B  
#GodBlessHolyRussia ☩",  
  "translator_type": "none",  
  "protected": false,
```



```
  "utc_offset": null,  
  "time_zone": null,  
  "geo_enabled": false,  
  "lang": null,  
  "contributors_enabled": false,
```

Geo-location of a tweet

- Geo-tagging information
 - highly accurate

```
{'_id': '1409920493596024845', 'date': datetime.datetime(2021, 6, 29, 18, 3, 35),  
'username': 'BigMarcs_',  
'text': '🇩🇪\U000e0067\U000e0062\U000e0065\U000e006e\U000e0067\U000e007f vs  
🇩🇪 COME ON ENGLAND!! 🤔 @ Wembley Stadium connected by EE  
https://t.co/KxaZBSgOam,
```

```
'geo_enabled': True, 'coordinates': [-0.27955033, 51.55598294],  
'location': 'England',  
'place_name': 'Brent, London',  
'place_country': 'United Kingdom', 'country_code': 'GB',  
'place_coordinates':  
[[[-0.336055, 51.527592], [-0.336055, 51.602799], [-0.190226, 51.602799], [-0.190226, 51.527592]]],  
'hashtags': [], 'mentions': [], 'source': '<a href="http://instagram.com" rel="nofollow">Instagram</a>'}
```



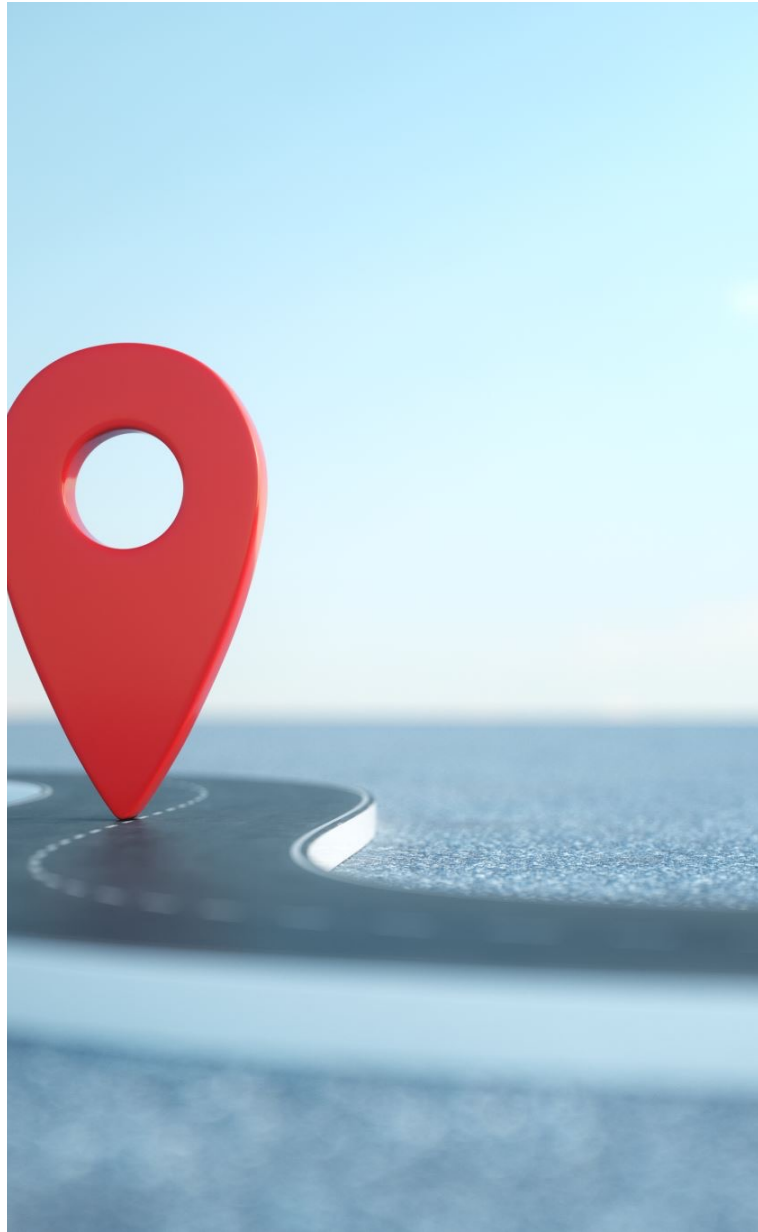
Place object

- A larger location
- {'_id': '1409603779373240328', 'date': datetime.datetime(2021, 6, 28, 21, 5, 4), 'username': 'lochdara', 'text': '@RoisinTCDNeuro It was a pleasure .', 'geoenabled': True, 'coordinates': None, 'location': 'Wexford, Ireland', 'place_name': 'Wexford, Ireland', 'place_country': 'Ireland', 'country_code': 'IE', 'place_coordinates': [[[-7.017507, 52.122381], [-7.017507, 52.797086], [-6.141269, 52.797086], [-6.141269, 52.122381]]], 'hashtags': [], 'mentions': ['RoisinTCDNeuro'], 'source': 'Twitter for Android'}
- {'_id': '1409603779767504896', 'date': datetime.datetime(2021, 6, 28, 21, 5, 4), 'username': 'debsknig', 'text': '@TBanton18 is playing a lovely innings this evening, so good to see him going well 57 from 28 so far #T20Blast2021', 'geoenabled': True, 'coordinates': None, 'location': 'Essex', 'place_name': 'Chelmsford, East', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[0.435659, 51.698636], [0.435659, 51.78115], [0.521299, 51.78115], [0.521299, 51.698636]]], 'hashtags': ['T20Blast2021'], 'mentions': ['TBanton18'], 'source': 'Twitter for iPhone'}

Geo-location of a tweet Place Object ?

- Derived from other info?

```
{ "user": { "derived": { "locations": [ { "country": "United States", "country_code": "US",  
  "locality": "Birmingham", "region": "Alabama", "sub_region": "Jefferson County",  
  "full_name": "Birmingham, Alabama, United States", "geo": { "coordinates":  
    [ -86.80249, 33.52066 ], "type": "point" } } ] } } }
```



Accuracy of Geo-tagging

- Geo-tagging gives GPS coordinates
 - How accurate are they?
 - Changes to the air in the upper atmosphere and ionosphere can also affect **accuracy**.
 - Neither a **GPS** or a smartphone will give you an exact position.
 - At best, you'll get an **accuracy** of 10 to 15 feet (3 to 5 meters).
 - Obstructions can lead to an **accuracy** of 5 to 30 feet (6 to 10 meters) or worse.

Issues

Geo-tagging

- Location enabled – accurate
- Geo-graphical area from profile – inaccurate

~1-5% of tweets are geo-tagged

- Location based data
 - New York city and Los Angels have 29% GPS-tagged

idea

- Use geo-annotated data
- Develop models to allocate geo-location
- Apply the model on non-geo-tagged tweets
- Assumption –
 - Textual information/clues
 - Non-geo tagged tweets and
 - Geo-tagged tweets follow same distribution
- Applicable in other scenarios too!

Part 1-

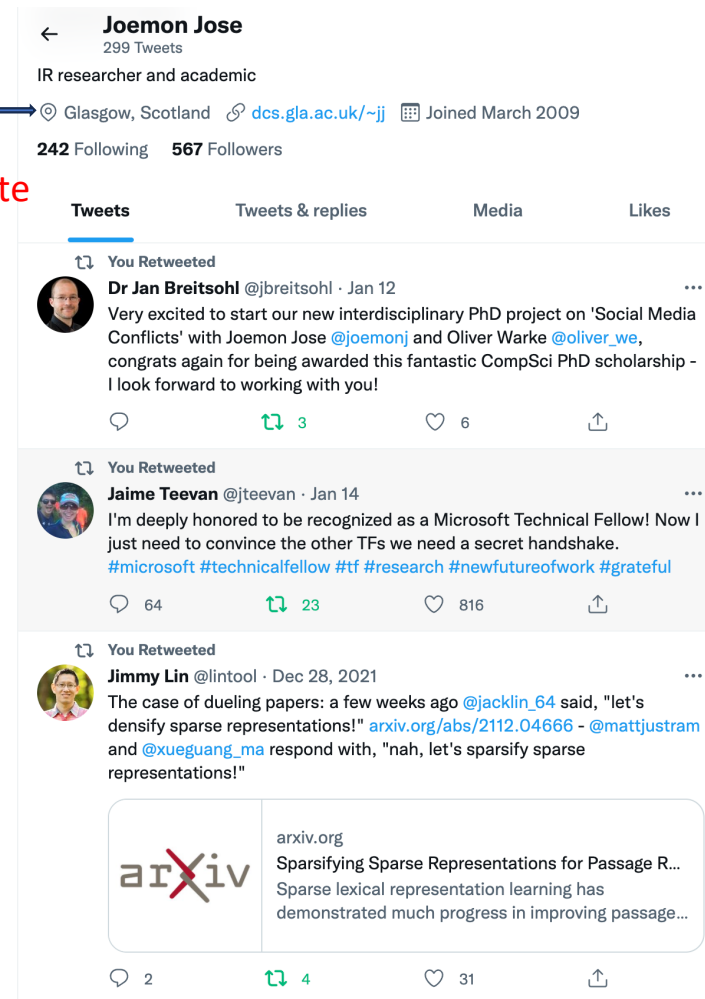
Approximately 1-5% of tweets published on the
twitter is geo-tagged

If we know the location of a tweet,

- Better understanding of an urban area
 - Tracking the way virus spreads
 - The identification of people that need help in case of disaster
 - better understanding the dynamics of a major event
- mainly to gather **Actionable insights!**
 - Applications and requirements
 - Transport tweets vs. financial tweets
- Two cases
 - Location of the user,
 - Where the user is based
 - Glasgow, west-end!
 - Benefits - marketing
 - Location of the tweet
 - From where a particular tweet was posted
 - Edinburgh Waverly station!
 - Benefits – what is happening?

User send the following tweets – where is his location?

If not, can we estimate from tweets send?



User location estimation

- Benefits
 - Marketing ...
- Issues
 - Privacy violation

Focusing on Tweets!

Country/State/Region Level (Coarse grained)

- Definition
 - Coarse-grained localisation
 - Trying to predict to a region level, city level etc.



Eric Holthaus ✓ @EricHolthaus · Jun 24

I'm not sure how to process this.

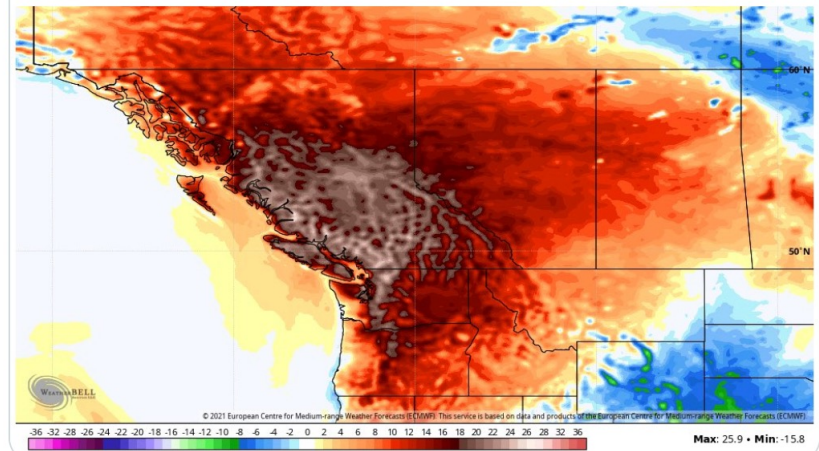
The normal late June high in **Vancouver** BC is 21°C (70°F). This **heat wave** will be 25°C (45°F) hotter than that — the equivalent cold wave would mean weather below freezing, colder than **Vancouver's** normal January.

This is an entirely new climate.



Tyler Hamilton @50ShadesofVan · Jun 24

An unfathomable temperature anomaly: approaching 25°C above normal for B.C. - huge red flag to have anomaly greater than average temperature. An equivalent negative anomaly would see highs near freezing. #BCHeat



Fine-grained localization

Text Clues ...

- Fine-grained localisation
 - Try to infer the location to street level

news_finale :----- : London fire: Huge FIREBALL 'explodes' under **Elephant & Castle station**, <https://t.co/DEYgxzuWTt>

ferozwala :----- : LIVE | Video from **#London's Elephant & Castle station**, where huge fire rages **#SputnikLive** **#Breaking**
<https://t.co/gkt94oRac5> via @SputnikInt

2021-06-28 15:21:55

2021-06-28 15:24:14





Given a tweet!

Where is its location?

Transportation

- Why is it needed?
 - Transportation, security
 - Event detection, event tracking



M25 crash near M40 junction closes two lanes causing long delays - recap

applications

Topic detection

- journalism

Disaster and emergency response

- Security and emergency services

Discuss what type of applications are possible with geo-localisation!

A cluster with four tweets!



Regional Interest in the topic

['changing', 'https://t.co/9ekXqXLUeq', 'invited', 'keep', 'landlords', 'rules', 'webinar']
['changing', 'https://t.co/siO3uXr0xE', 'invited', 'keep', 'landlords', 'rules', 'webinar']
['changing', 'https://t.co/tR7Q7VWf5e', 'invited', 'keep', 'landlords', 'rules', 'webinar']
['changing', 'https://t.co/eB0b8EDvZM', 'invited', 'keep', 'landlords', 'rules', 'webinar']

Tweetids
1351844052941959170
1351844052950425601
1351844053923491840
1351844054317686786

Fine grained vs coarse grained

- Definition
 - Coarse-grained localisation
 - Trying to predict to a region level, city level etc.
 - Example cluster (coarse grained)
 - Northeast England
 - Newcastle
 - Breckland Council, Council in Dereham, England



Fine-grained localisation

- Fine-grained localisation
 - Try to infer the location to street level

news_finale :----- : London fire: Huge FIREBALL 'explodes' ***under Elephant & Castle station***

<https://t.co/DEYgxzuWTt>

ferozwala :----- : LIVE | Video from [#London's Elephant & Castle station](#), where huge fire rages

#SputnikLive #Breaking

<https://t.co/gkt94oRac5> via @SputnikInt

Textual
Clues!!!

2021-06-28 15:21:55

2021-06-28 15:24:14

Location

London Rd, Elephant and Castle, London SE1 6LW

Elephant & Castle is a London Underground station in the London Borough of Southwark in south London. It is on the Bank branch of the Northern line between Kennington and Borough stations

?

- **Euston road**
traffic accident
- Road Traffic
Accident **Central
London**
Motorcyclist
injured and being
treated by medics



Different clues emerge as different people tweets!

Problem statement

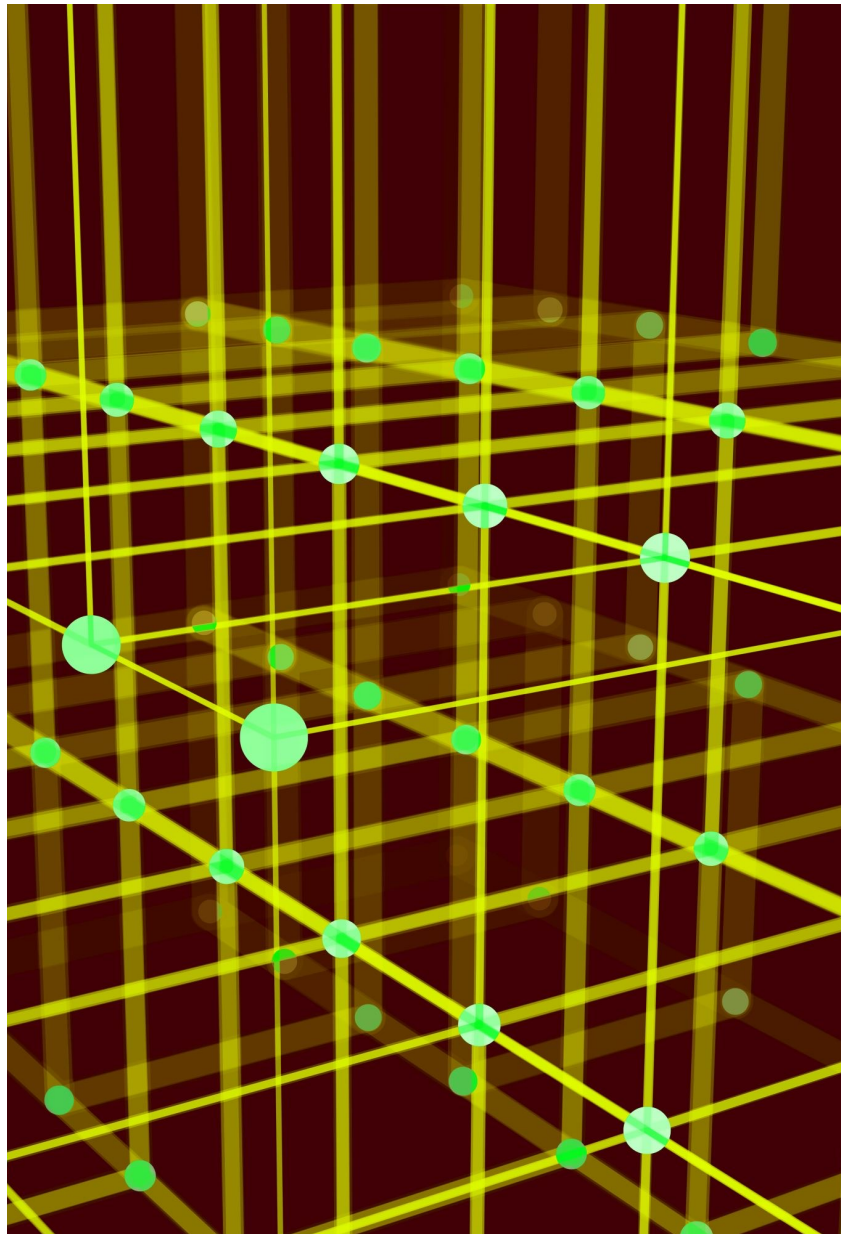
- Geo-localisation
 - Provide location estimates for individual tweets
 - Coarse-grained Geo-localisation
 - Provide location estimates for individual tweets at regional or country level
 - fine-grained Geo-localisation
 - Provide location estimates for individual tweets at city or neighbourhood level
- Approach
 - Train a model on a geo-tagged data set
 - Validate and test on geo-tagged data
 - Test on non-geo tagged data as well

summary

- We have looked into geo-localisation
 - ***Two (Three) forms of it***
- We have visited the various forms of location data encoded in a tweet
 - Location?
 - GPS tags/geo coordinates
 - Place_object ?
- We have looked into various applications

```
in_reply_to_screen_name : null,  
"user": {  
  "id": 3131372879,  
  "id_str": "3131372879",  
  "name": "Uncle Memester",  
  "screen_name": "UncleMemester",  
  "location": "Matthew 24:13",  
  "url": null,  
  "description": "Parody! Fluent in #GenX. B  
#GodBlessHolyRussia †",  
  "translator_type": "none",  
  "protected": false,
```





Part II

Grid based Geo-localisation



Wellman cars @WellmanCars · Mar 31, 2019

Accident Glasgow road,Blantyre road closed due to overturned car.
Between the Parkville and the **west end** bar.



Geo-coordinates of a tweet?

- Given a tweet
 - What would be its location?
- In fact, we see a tweet without any coordinate information
 - Can we predict its location
 - If you can how close to its actual location

Location clues!



Wellman cars @WellmanCars · Mar 31, 2019



Accident Glasgow road, Blantyre road closed due to overturned car.
Between the Parkville and the **west end** bar.



Given a tweet

- Search for similar tweets
 - As in Google, Bing etc
 - For example, *“Accident Glasgow road, Blantyre road closed due to overturned car. Between the Parkville and the west end bar.”*
- Then we will assign location of its closest tweet(s)
- Idea
 - Create query strings
 - Search BUT where?

Idea - search



Google

Accident Glasgow road, Blantyre road closed due to overturned car. B X

News Images Videos Books Finance

About 430 results (0.55 seconds)

Facebook · Blantyre Telegraph
100+ reactions · 4 years ago

Blantyre Telegraph - ACCIDENT: GLASGOW ROAD Traffic ...
car has overturned on Glasgow Road between the. Parkville and West End. The Road is currently closed. Emergency services are in attendance ...
Missing: bar | Show results with: bar.

Daily Record
https://www.dailyrecord.co.uk › ... › Police Scotland

Blantyre road closed as emergency services rush to multi ...
19 Mar 2018 — Blantyre's Glasgow Road has been closed this afternoon following a multi-vehicle accident. Emergency services were called to the scene, ...

Facebook · Blantyre Telegraph
60+ reactions

GLASGOW ROAD, BLANTYRE Earlier Vehicle Accident ...
vehicle road crash on Glasgow Road, Blantyre." The road was closed off, with a large police presence until late in the afternoon. An ambulance took away one ...
Missing: overturned Parkville

Glasgow Times
https://www.glasgowtimes.co.uk › news › scottish-news

'Avoid' warning as Glasgow Road closed after police incident
3 Aug 2023 — Greater Glasgow Road Policing said on Twitter: "Glasgow Road in Blantyre is currently closed following a road crash involving one vehicle.
Missing: overturned Parkville

STV News
https://news.stv.tv › Glasgow & West

Drivers told to 'avoid' Blantyre's Glasgow Road area as ...
3 Aug 2023 — Drivers in South Lanarkshire are being urged to avoid a major road as emergency crews attend a crash. Crews were called to Glasgow Road, ...
Missing: overturned Parkville end bar.

Glasgow Times
https://www.glasgowtimes.co.uk › news › scottish-news

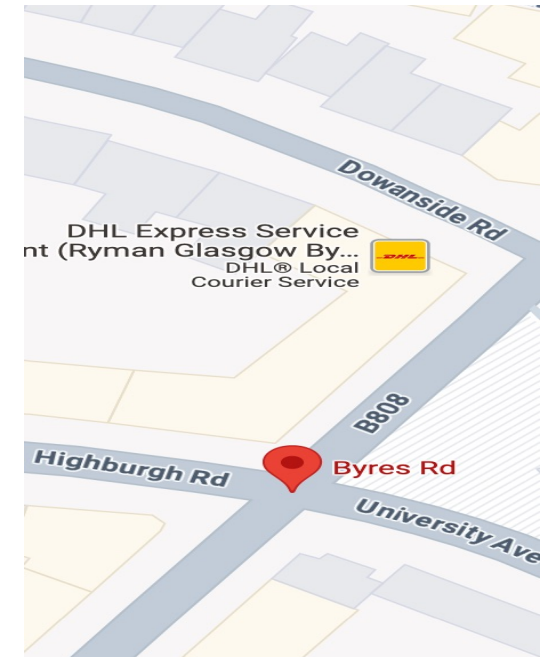
Person hit by lorry near shopping centre as First Bus ...
19 Dec 2023 — A person has been knocked down by a lorry near a shopping centre. Emergency services raced to Glasgow Road near Clydeview Shopping Centre, ...
Missing: overturned Parkville

Remember - Tweets are

- Short text
 - Maximum 280 characters
 - *More in certain situations under new management!*
- Informal communication
- Unstructured
- Quality of text in terms of syntax and semantics
- Use of emojis
- Noisy nature
 - Use of colloquial forms of words
 - Spelling mistakes

Proposal is that!

- Group tweets from an area/sub-area
 - Create a larger document
 - Why?
- Create an archive of past documents
 - Tweets from that location
 - Grouped tweets from each location
- Create a retrieval system
 - Input – tweet text or selected tokens/terms
 - Output – ranked list of grouped documents!!



Tweets from these area may have words like

- *Little Italy*
- *Tindex box*
- *Rental services*
- *Spartini*
-

Grouping Tweets from Byres Road instead of.....

- Search a collection of such documents (instead of Google/Bing!)
- Search for similar tweets
 - For example, *“Accident Glasgow road, Blantyre road closed due to overturned car. Between the Parkville and the west end bar.”*
- Then we will assign location of its closest tweet(s)
- Idea
 - Create query strings
 - Search BUT where?



Google

Accident Glasgow road, Blantyre road closed due to overturned car. B X

News Images Videos Books Finance

About 430 results (0.55 seconds)

Facebook · Blantyre Telegraph
100+ reactions · 4 years ago

Blantyre Telegraph - ACCIDENT: GLASGOW ROAD Traffic ...
car has overturned on Glasgow Road between the. Parkville and West End. The Road is currently closed. Emergency services are in attendance ...
Missing: bar · Show results with: bar.

Daily Record
<https://www.dailyrecord.co.uk> · ... · Police Scotland

Blantyre road closed as emergency services rush to multi ...
19 Mar 2018 — Blantyre's Glasgow Road has been closed this afternoon following a multi-vehicle accident. Emergency services were called to the scene, ...

Facebook · Blantyre Telegraph
60+ reactions

GLASGOW ROAD, BLANTYRE Earlier Vehicle Accident ...
vehicle road crash on Glasgow Road, Blantyre." The road was closed off, with a large police presence until late in the afternoon. An ambulance took away one ...
Missing: overturned Parkville

Glasgow Times
<https://www.glasgowtimes.co.uk> · news · scottish-news

'Avoid' warning as Glasgow Road closed after police incident
3 Aug 2023 — Greater Glasgow Road Policing said on Twitter: "Glasgow Road in Blantyre is currently closed following a road crash involving one vehicle.
Missing: overturned Parkville

STV News
<https://news.stv.tv> · Glasgow & West

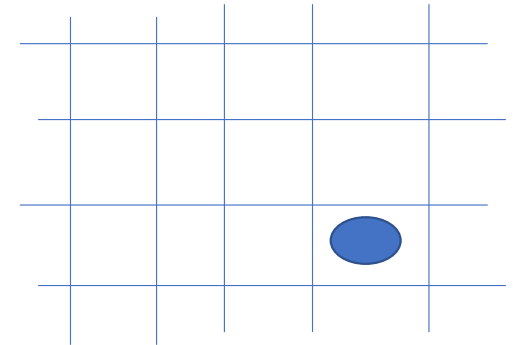
Drivers told to 'avoid' Blantyre's Glasgow Road area as ...
3 Aug 2023 — Drivers in South Lanarkshire are being urged to avoid a major road as emergency crews attend a crash. Crews were called to Glasgow Road, ...
Missing: overturned Parkville end bar.

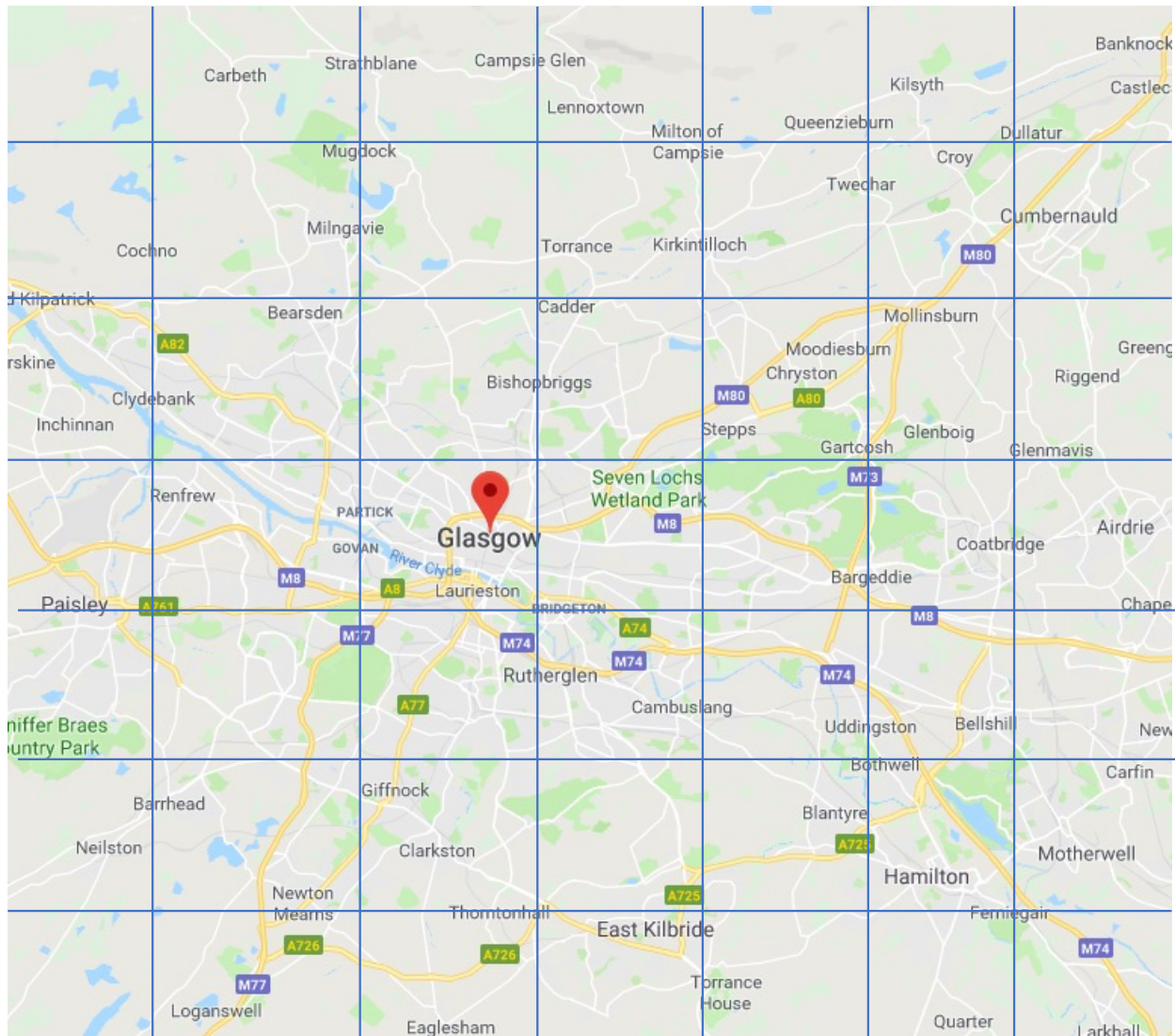
Glasgow Times
<https://www.glasgowtimes.co.uk> · news · scottish-news

Person hit by lorry near shopping centre as First Bus ...
19 Dec 2023 — A person has been knocked down by a lorry near a shopping centre. Emergency services raced to Glasgow Road near Clydeview Shopping Centre, ...
Missing: overturned Parkville

Grid based approach – Idea!

- Divide geographical space
 - areas/grids
- Represent tweets in those area as a concatenated vector
 - Consider it as a document
 - Vector representation?
 - How to make a vector representation?
- Given a tweet to localise, rank the grouped tweets
- Assign location of the most similar area





London

- Geo area

$[-0.563, 51.261318,$
 $0.28036, 51.686031]$

Rows

59

Columns

48

Grid 1 to 2832

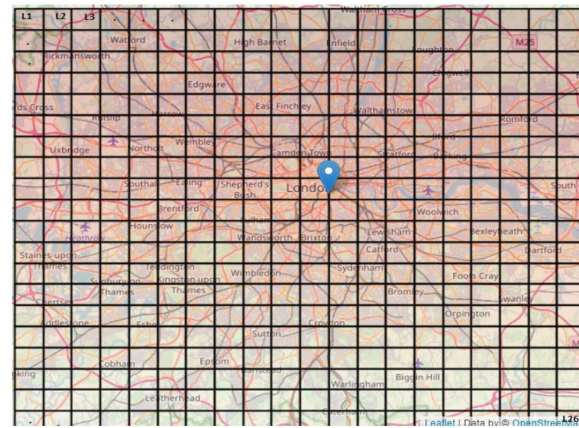
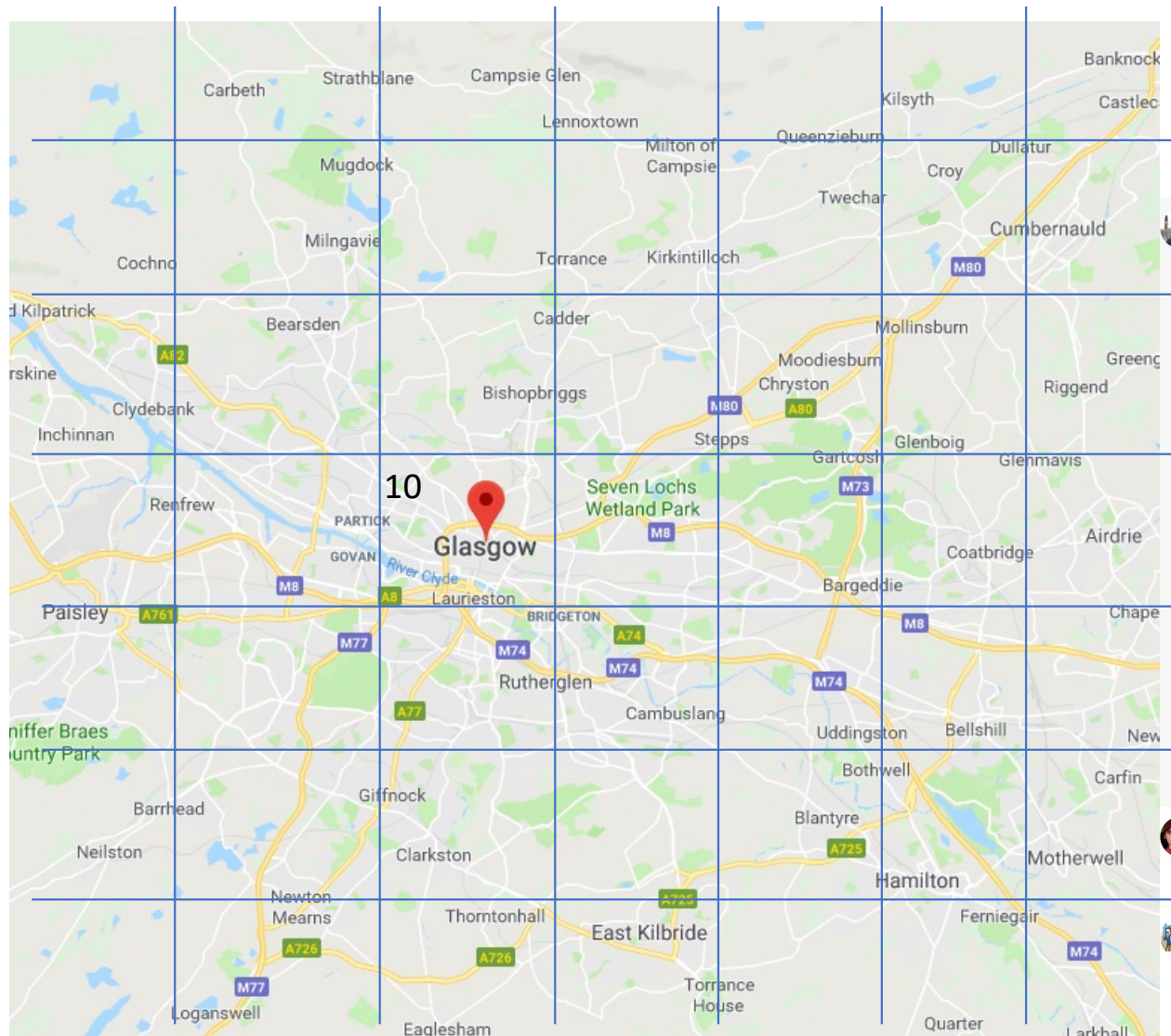


Fig 12: Diagrammatic representation of dividing London into 1Km2 Grids



Paul Sweeney @PaulSweeney · Jan 22

The Hilton Grosvenor at the top of Byres Road today might look identical to this 1931 photo, but the sandstone building was actually demolished and completely rebuilt as a faithful replica in moulded concrete after the disastrous fire of January 1978, during the firemen's strike.



Glasgow City Archives @GC_Archives · Jan 22

Terraced houses, 1 Grosvenor Terr, Dowanhill, 1931
Designed by J T Rothead, the Venetian-style terrace was completed 1858. The Grosvenor Hotel opened at Nos 1 and 2 in 1938. It was extended in the 1970s. It reopened in 1982 after fire in 1978. ...

8

5

72



Greg @MedicalTango · Jan 22

they should enforce a no english people on byres road rule for 1 day just to see what it looks like

3

4

55



Mémoire2Ville @Memoire2cite · 22h

Byres Road/Highburgh Road junction, Glasgow. 1996

Vector representation

- MP Kerry McCarthy: 'We Need Action Not Words On #AnimalWelfare'
<https://t.co/QgmG6rxvfq>
 - ['#AnimalWelfare', 'action', 'https://t.co/QgmG6rxvfq', 'kerry', 'mccarthy', 'need', 'words', 'we']
- @LissaKEvans Do no different from the last labour government. Also I hate the Toxic British Press and TV news channels. They don't get to choose who runs our country. We the ppl do. I've no love for Boris but he dot Brexit done and I voted leave. The rest is normal and will never change!
 - ['@LissaKEvans', 'boris', 'brexit', 'british', 'change', 'channels', 'choose', 'country', 'different', 'done', 'don't', 'dot', 'government', 'hate', 'i've', 'labour', 'leave', 'love', 'never', 'news', 'normal', 'ppl', 'press', 'rest', 'runs', 'toxic', 'voted']
- ok horny as f**k need sex in uk dont care if man or woman
 - ['care', 'dont', 'horny', 'man', 'need', 'sex', 'woman']

Vector representation for a tweet & weighting

- For each tweet,
- We create a vector representation
 - ['#ReadForEmpathy', 'books', 'brilliant', 'excellent', 'excellent', 'idea', 'utterly']
 - 7 terms
- Term weighting
 - **TF – term frequency**
 - How many times a term appeared in a document?
 - **Inverse document frequency (idf)**
 - $\text{Log}(N/n)$;
 - N = total number of documents in the collection
 - n – no of documents in which a term appears
 - $\text{TF} * \text{idf}$

Weighting

- TF approximation?
- Every term get equal weights?
 - For example $1/\sqrt{7}$?
 - ['#ReadForEmpathy', 'books', 'brilliant', 'excellent', 'excellent', 'idea', 'utterly']
- Idf -?
 - We will analyse the collection and to compute this value
 - Group all documents in our database
 - Total number of documents
 - No of documents in which a particular term appears

Vector representation of a grid

- ['#ReadForEmpathy', 'books', 'brilliant', 'excellent', 'idea', 'utterly']
 - Each term has a weight, $1/\sqrt{6}$
- ['care', 'dont', 'horny', 'man', 'need', 'sex', 'woman']
 - Each term has a weight of $1/\sqrt{7}$
-
- Concatenate representation of all tweets,
 - ['#ReadForEmpathy', 'books', 'brilliant', 'excellent', 'idea', 'utterly'] + ['care', 'dont', 'horny', 'man', 'need', 'sex', 'woman']
 - ['#ReadForEmpathy', 'books', 'brilliant', 'care', 'dont', 'excellent', 'horny', 'idea', 'man', 'need', 'sex', 'woman', 'utterly']

Part III

Distance between points

Euclidean distance between two points

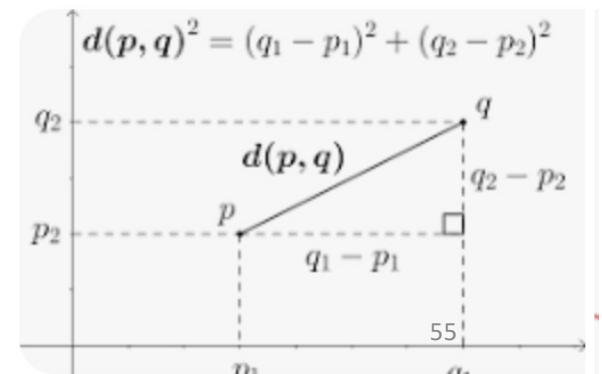
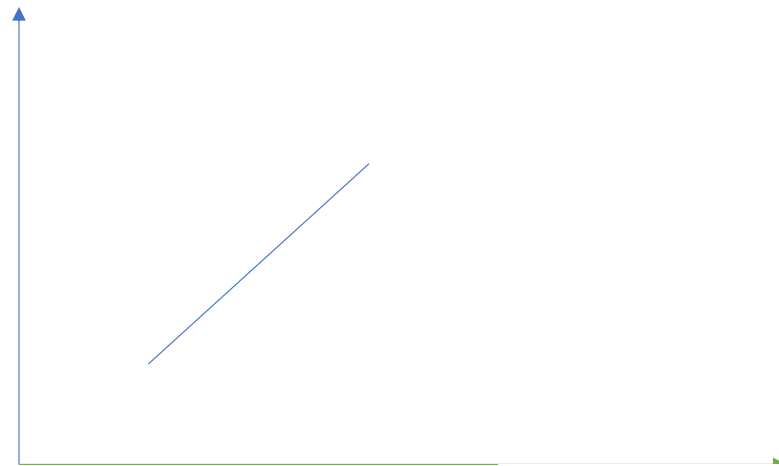
- (p_1, p_2)
- (q_1, q_2)

$$d(\mathbf{p}, \mathbf{q}) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

$\mathbf{p}, \mathbf{q}$ = two points in Euclidean n -space

q_i, p_i = Euclidean vectors, starting from the origin of the space (initial point)

n = n -space



earth

- Spherical surface



Distance between two points on earth

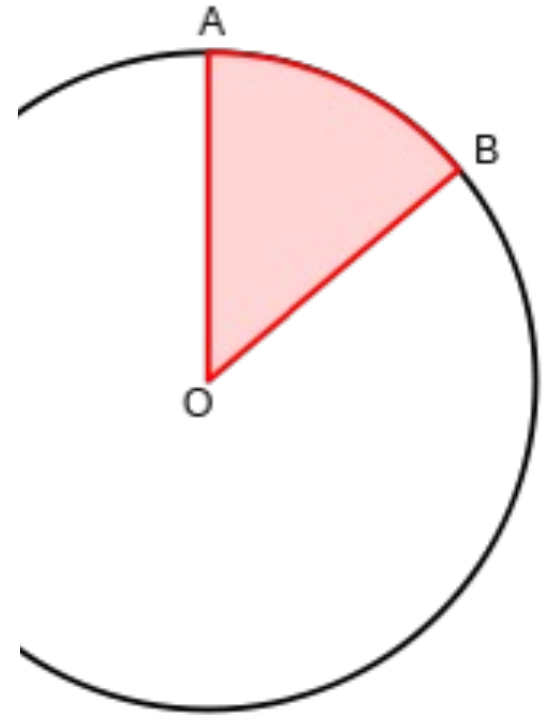
- You need the shortest path on the surface of the Earth, which is roughly spherical.
- On a sphere the shortest path between two points is along an arc of a *great circle*:
 - circle drawn on the surface of the sphere which is centred on the same point as the sphere
 - and has the same radius.
- Any two points lie on a unique great circle, which they divide up into two arcs.
- The shortest path between the points is along the shorter of these two arcs.



Haversine formula

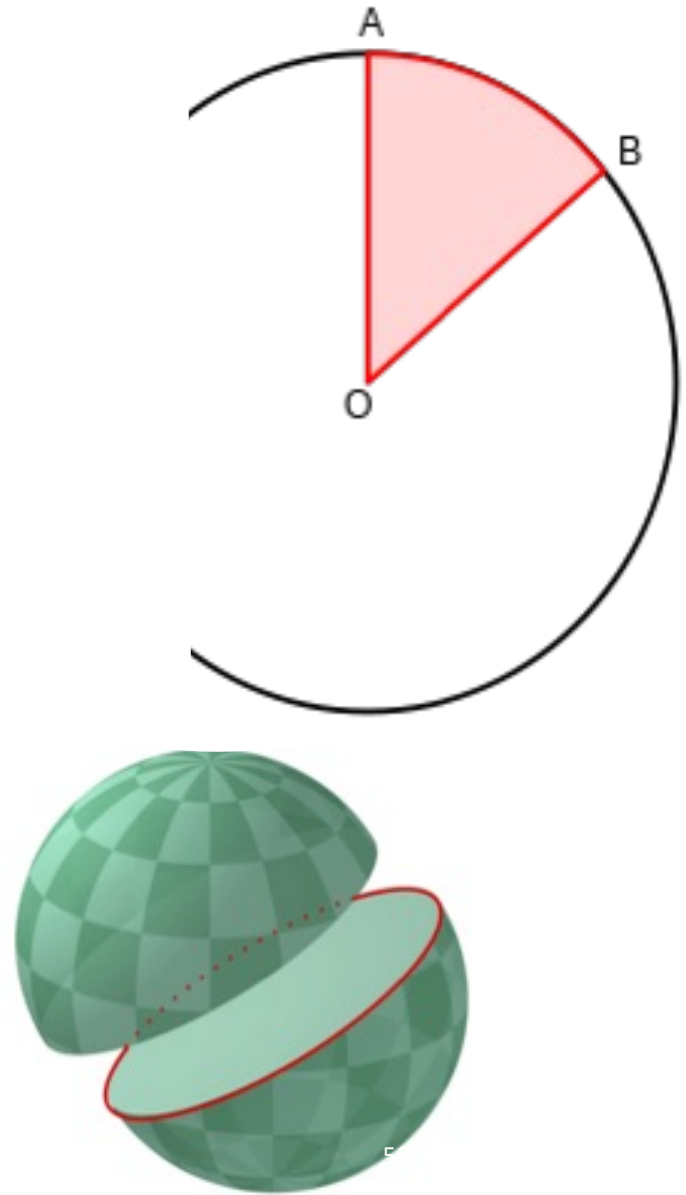
- Let the central angle Θ between any two points on a sphere

$$\text{haversin}(\theta) = \sin^2(\theta/2).$$



Haversine formula

- A great circle divides the sphere in two equal hemispheres
- d is the distance between the two points along a great circle of the sphere
- r is the radius of the sphere.
 - Radius of the earth



Haversine distance

Haversine $a = \sin^2(\Delta\phi/2) + \cos \phi_1 \cdot \cos \phi_2 \cdot \sin^2(\Delta\lambda/2)$

formula: $c = 2 \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a})$

$d = R \cdot c$

where ϕ is latitude, λ is longitude, R is earth's radius (mean radius = 6,371km);

note that angles need to be in radians to pass to trig functions!

```
JavaScript: const R = 6371e3; // metres
const  $\phi_1$  = lat1 * Math.PI/180; //  $\phi$ ,  $\lambda$  in radians
const  $\phi_2$  = lat2 * Math.PI/180;
const  $\Delta\phi$  = (lat2-lat1) * Math.PI/180;
const  $\Delta\lambda$  = (lon2-lon1) * Math.PI/180;

const a = Math.sin( $\Delta\phi/2$ ) * Math.sin( $\Delta\phi/2$ ) +
          Math.cos( $\phi_1$ ) * Math.cos( $\phi_2$ ) *
          Math.sin( $\Delta\lambda/2$ ) * Math.sin( $\Delta\lambda/2$ );
const c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1-a));

const d = R * c; // in metres
```

In Python

```
def computeDistance(self, long2, lat2):  
    # Haversine Formula  
    # a = math.sin(dlat / 2)**2 + math.cos(lat1) * math.cos(lat2) * math.sin(dlon / 2)**2  
    # c = 2 * math.atan2(math.sqrt(a), math.sqrt(1 - a))  
    # distance = R * c  
    R = 6373.0  
    lat1 = self.boundingCoordinates[1]  
    long1 = self.boundingCoordinates[0]  
    lat2 = self.boundingCoordinates[3]  
    lon2 = self.boundingCoordinates[2]  
  
    phi1 = lat1 * (math.pi / 180)  
    phi2 = lat2 * (math.pi / 180)  
    # Delta Phi  
    delta1 = (lat2-lat1) * (math.pi / 180)  
    # Delta Lambda  
    delta2 = (long2-long1) * (math.pi / 180)  
  
    a= math.sin(delta1 / 2) ** 2 + math.cos(phi1) * math.cos(phi2) * math.sin(delta2/2) ** 2  
    c= 2 * math.atan2(math.sqrt(a), math.sqrt(1 - a))  
    d= R * c  
    # print(' the distnace between points is : ', d)  
    return d
```

Diagram illustrating the coordinate mapping for the Haversine formula:

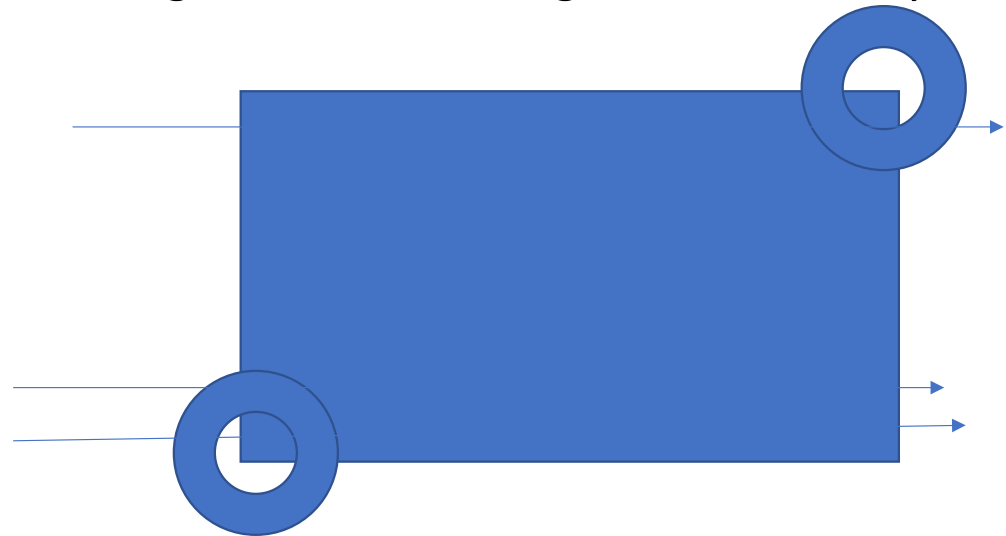
- The boundingCoordinates list is indexed as follows:
 - `lat1` is `self.boundingCoordinates[1]`
 - `long1` is `self.boundingCoordinates[0]`
 - `lat2` is `self.boundingCoordinates[3]`
 - `lon2` is `self.boundingCoordinates[2]`
- The `phi` values are calculated as `phi1 = lat1 * (math.pi / 180)` and `phi2 = lat2 * (math.pi / 180)`.
- The `delta` values are calculated as `delta1 = (lat2-lat1) * (math.pi / 180)` and `delta2 = (long2-long1) * (math.pi / 180)`.

Visual representation of the coordinate indices:

- A blue rectangle represents the bounding box coordinates, with a label `[(2), [3)]` indicating the range of indices used for `lat2` and `lon2`.
- A white box labeled `[(0], [1)]` indicates the range of indices used for `long1` and `lat1`.

Geographic Grid Allocation for Tweets

- the given location is divided into square grids of area 1km
- The grids are allocated a grid id starting from
 - L0 for the grid at the lower bounding latitude and longitude of the city to
 - Lmax for the grid at the upper bounding latitude and longitude of the city.



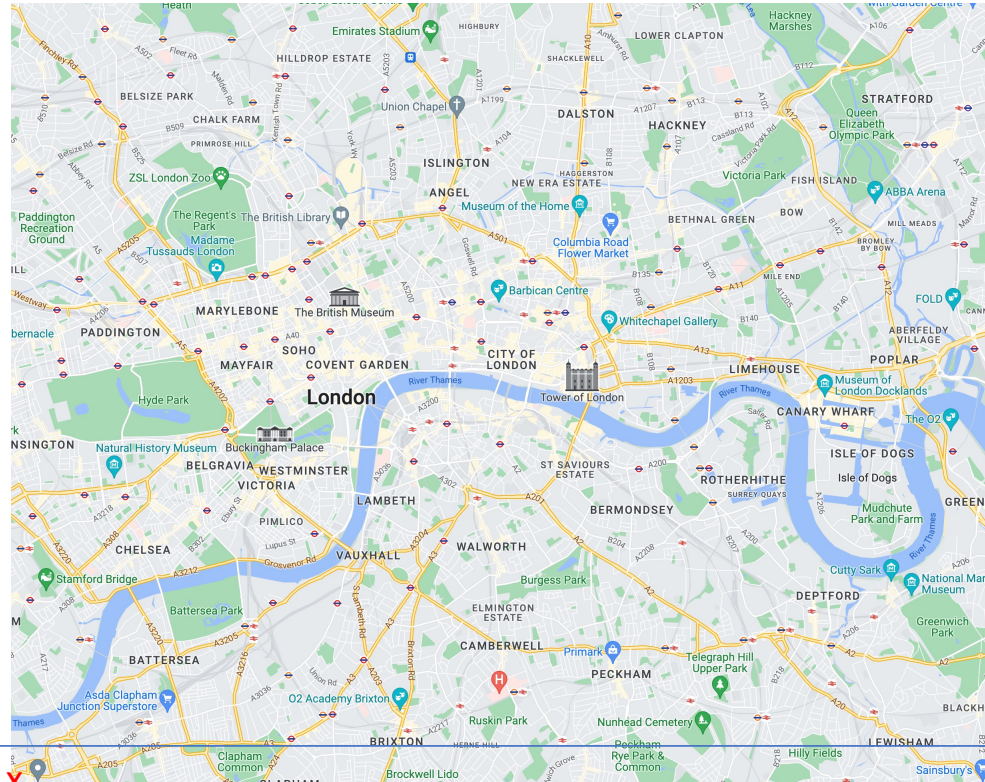
Task1 – course work

- (i) A dataset will be given to you (Teams File area). Write python code to organise tweets into grids of 1km x 1km. Draw charts and/or figures to analyse the distribution of data.
- The coordinate system we used to collect data is
 - London = [-0.563, 51.261318, 0.28036, 51.686031]

London – boundary box

(0.28036, 51.686031)

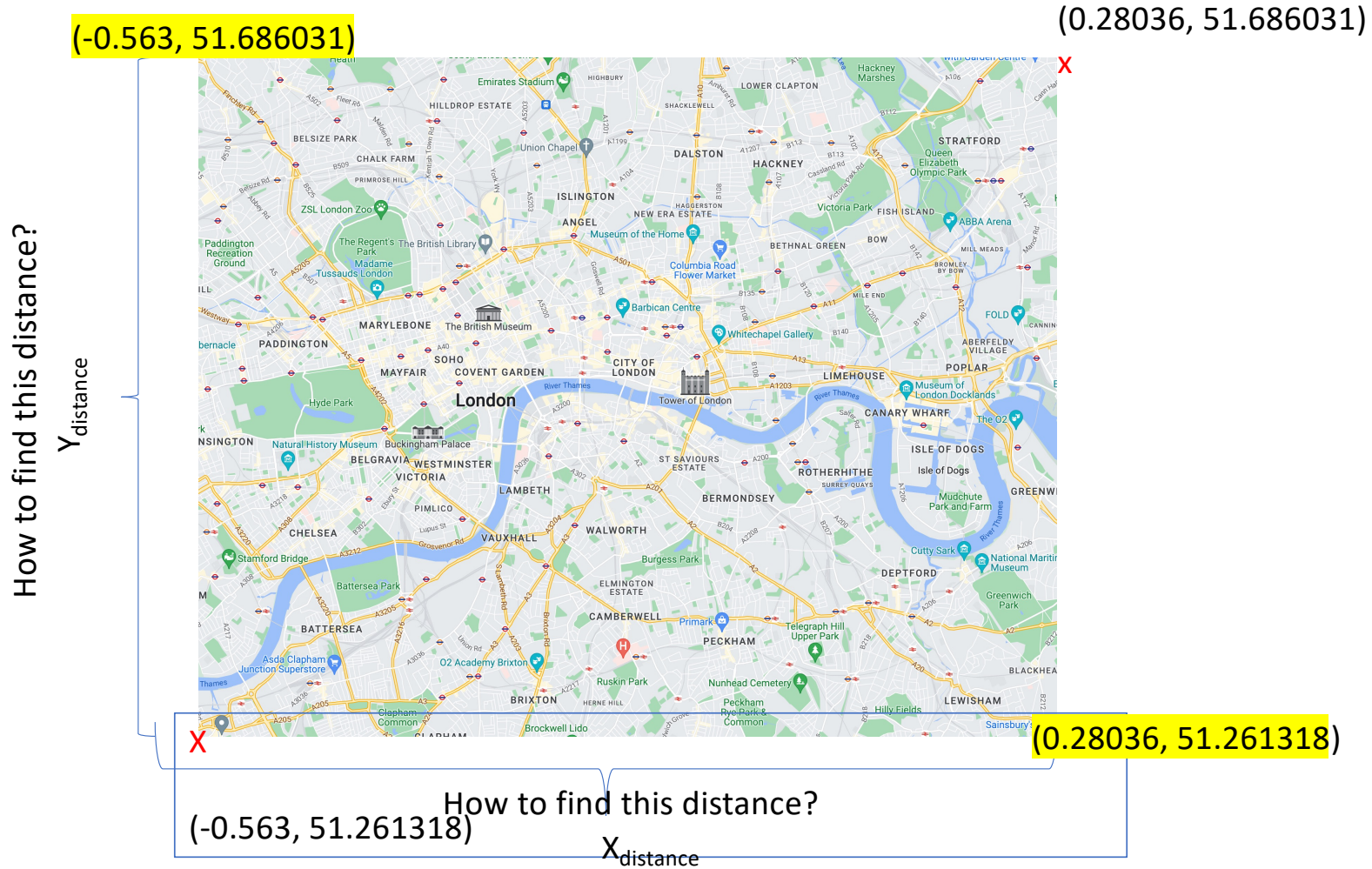
How to find this distance?



(-0.563, 51.261318)

How to find this distance?

London – boundary box



Horizontal and vertical distances

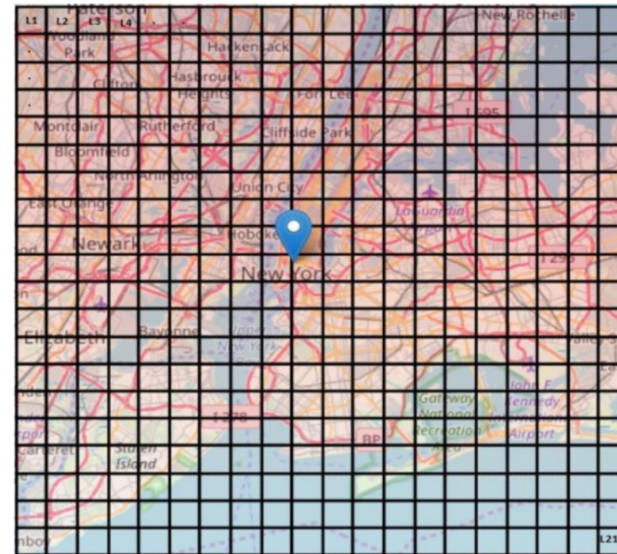
- X_{distance}
- Y_{distance}
- Closest Integer
 - They are roughly is the number of columns of distance 1KM
- rows =
`int(np.ceil(computeDistance(boundingCoordinates[0], boundingCoordinates[3])))`
- columns
`=int(np.ceil(computeDistance(boundingCoordinates[2], boundingCoordinates[1])))`

```
computeDistance(self, long1, lat1, long2, lat2):  
    # Haversine Formula  
    # a = math.sin(dlat / 2)**2 + math.cos(lat1) * math.cos(lat2) * math.sin(dlong / 2)**2  
    # c = 2 * math.atan2(math.sqrt(a), math.sqrt(1 - a))  
    # distance = R * c  
    R = 6373.0  
    lat1 = self.boundingCoordinates[1]  
    long1 = self.boundingCoordinates[0]  
    lat2 = self.boundingCoordinates[3]  
    lon2 = self.boundingCoordinates[2]  
  
    phi1 = lat1 * (math.pi / 180)  
    phi2 = lat2 * (math.pi / 180)  
    # Delta Phi  
    delta1 = (lat2-lat1) * (math.pi / 180)  
    # Delta Lambda  
    delta2 = (long2-long1) * (math.pi / 180)  
  
    a = math.sin(delta1 / 2) ** 2 + math.cos(phi1) * math.cos(phi2) * math.sin(delta2 / 2) ** 2  
    c = 2 * math.atan2(math.sqrt(a), math.sqrt(1 - a))  
    d = R * c  
    # print(' the distnace between points is : ', d)  
    return d
```

New York

[-74.255653,40.495682,-
73.689484,40.917622]

- Rows 48
- Columns 47
- Grids1 to 2256



Part IV

Geo-localisation technique – an updated idea
An information retrieval approach

Geo-localised tweet

- Tweets with geo-coordinates
- Collect lots of them from a city
- Focus on
 - Coordinates
 - text
- Consider a city
 - London
 - New York

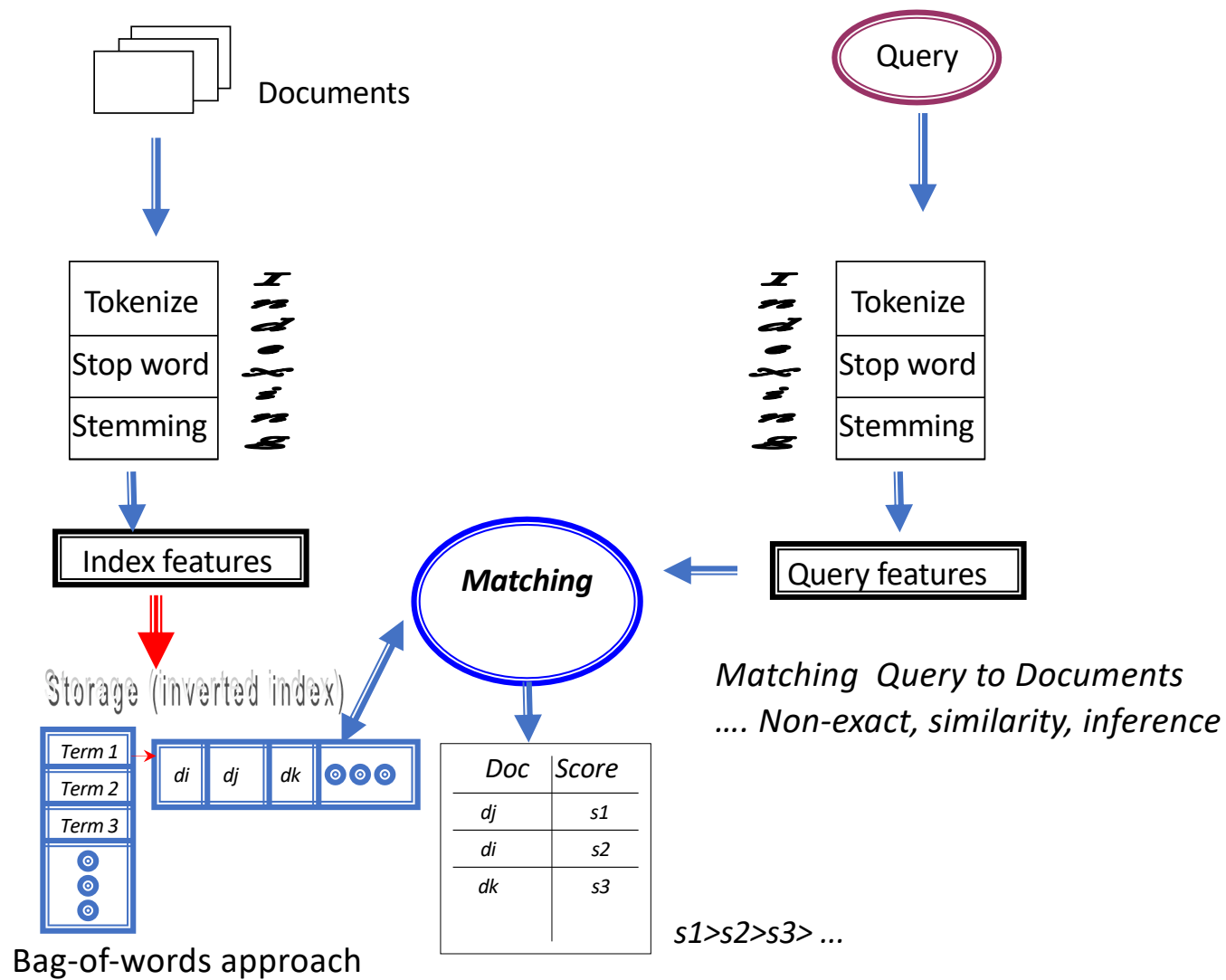
Overall approach – more concrete!

- Divide geographical area of interest into a grid of 1km squared areas and
- associate each geo-tagged tweet to an area based on its location
- Top-N content-based most similar geo-tagged tweet
 - **Use retrieval models/retrieval engine/ranking engine**
 - Bing/Google
- Assign top-ranked item

An information retrieval scheme

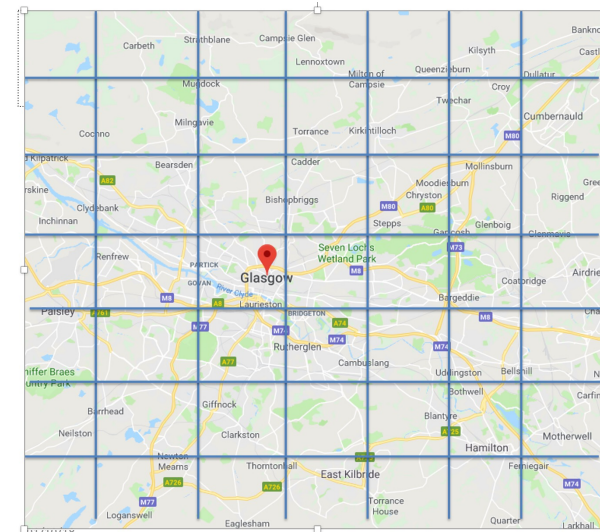
- Dividing geographical space
 - Into fine grained 1km grids
 - Aggregate all the tweets in a grid
 - Tweet 1 + Tweet 2+ Tweet 3+ +Tweet n
 - Tweet text only
 - May lose coordinates and other meta-data – time – user id – etc
- Lab task!

Write python code to organise tweets into grids of 1km x 1km.



When you get a new tweet

- Given a new tweet to geo-tag
 - ***End of an era. Happy retirement to a Glasgow legend.***
 - Find similar tweets
 - Top document and the associated grid
 - Assign that grid as tweets location
- Assuming you did the lab task
 - Find similarity between tweets



Some issues

- Group all tweets in a grid
 - Grouping tweets may lead to
 - Aggregation can be noisy
 - Infrequent but relevant geo-tagged tweets could get lost
 - Losing the semantics
 - Too many words outside the area
 - important evidence encoded in the meta-data is also lost?
- Eventually affect the retrieval quality
- Top-ranked documents
 - If the correct document is top second!

Granular approach

Refine our current idea further

Granular approach

- Instead of aggregating all the tweets in a grid, just consider all tweets individually

- That is, we don't do

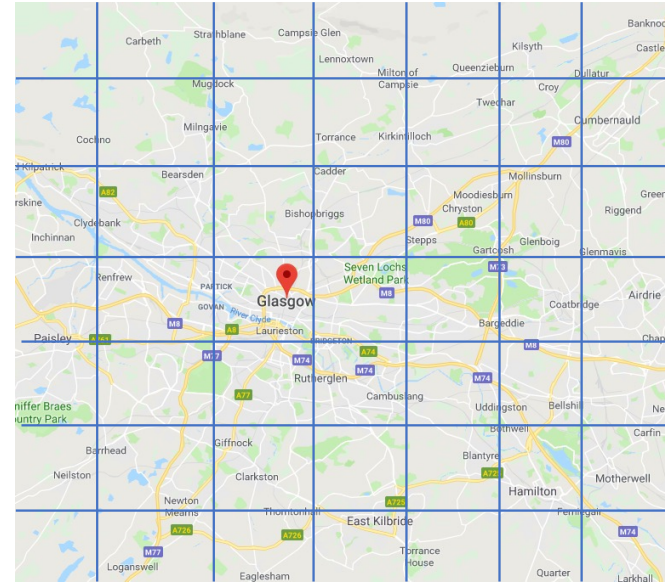
- ~~concatenate all the tweets in a grid~~

- ~~Tweet 1 + Tweet 2 + Tweet 3 + + Tweet n~~

May lose coordinates and other meta-data –
time – user id - etc

- We treat all tweets individually
- In the generic approach, we will have 100 docs for 10KM rectangular area, divided into 1KM
- In the granular approach we will have N documents
 - **N number of tweets in the system**

How many
documents in the
original idea?



80

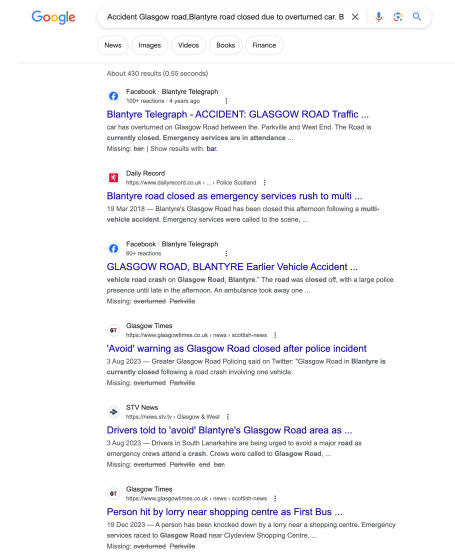
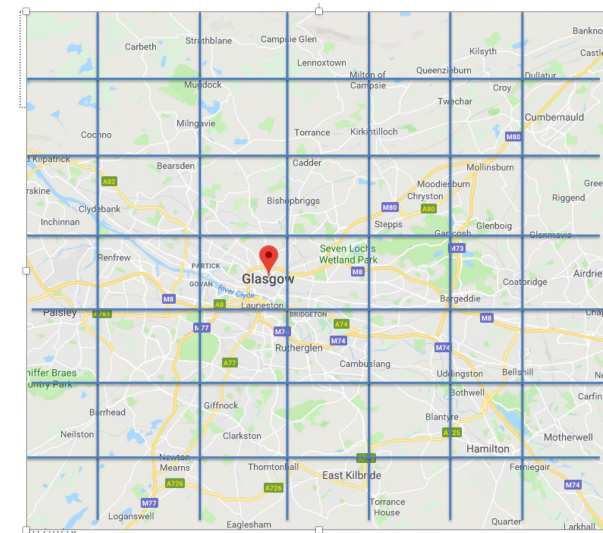
7x7 grid

Total 49 documents in this case!

Given a tweet

End of an era. Happy retirement to a New York legend.

- extract the text features
- ['end', 'era', 'happy', 'legend', 'new', 'retirement', 'york']
- Retrieve similar tweets
 - Not the document for a grid location !
- Consider top-k tweets
 - They are the most similar ones to the given tweet based on the content
 - Look at their geo-coordinates
 - If they all point to same grid, assign that grid centroid as tweets location
 - Otherwise
 - Just use the top ranked the tweet – assign its coordinates
 - Or ...

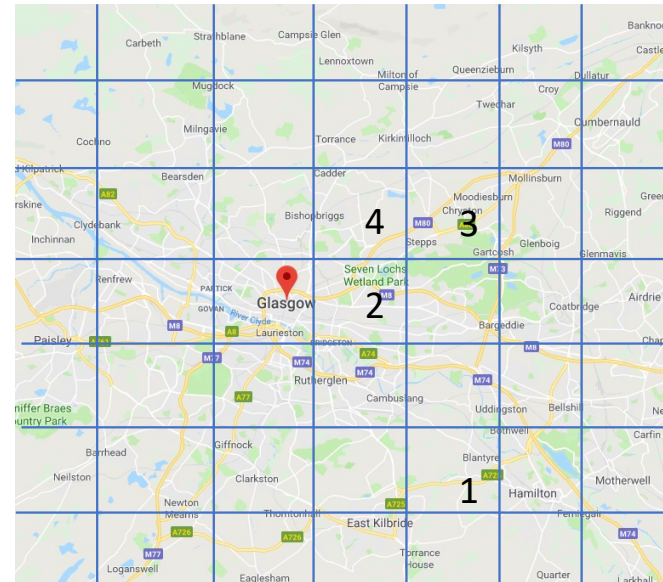


Voting algorithms

- If we have more than one potential locations as candidates,
- we can consider voting algorithms
 - **Majority voting**
 - Consider top-3 ranked docs
 - 2 from **grid id -10** ; 1 from **grid id - 13**
 - **We assign centroid of grid id -10**
 - In this case we have two strong indicators ...

Given a query
And a list of ranked tweets!!

Majority voting



83

“On fine-grained geolocalisation of tweets and real-time traffic incident detection”

Jorge David Gonzalez Paule, Yeran Sun, Yashar Moshfeghi

- <https://doi.org/10.1016/j.ipm.2018.03.011>

Weighting maybe qualitative as well
Weighted voting – using credibility

Credibility of tweets

- Can we apply credibility score to tweets?
 - Credibility of a source in time of emergency
 - To remove noises, rumors etc.
- Define credibility
 - The credibility of a user is score that represents user's posting activity and its relevance to the physical location they are posting from
 - ??
- methods to compute credibility
 - ~~• Linear regression models~~
 - ~~• Need to identify disambiguating features~~

Extracting credibility from a tweet's source

- For each tweet in the validation set, Top-N content-wise most similar tweet
 - From the training set
- Geographical distance between the tweet
 - In the validation set
 - Each element of its top-N ranked tweets
 - Credibility of s_i
- Each source s_i
 - Define TN_i
 - {all tweets appearing in any of top-N rankings (ts_i), produced for each element of the validations set tv_i }

Credibility distribution

- We define set TN_i
 - All tweets appearing in any of the the top_N rankings (t_{si}) for every tweet in the validation set

$$Credibility(s_i) = \frac{|\{t_{si} \in TN_i \mid distance(t_{si}, t_{vi}) \leq 1km\}|}{|TN_i|}$$

- Top_N content wise most similar tweets for every tweet in the validation set
- Geographical distance between the tweet in validation set and each element in its Top_N

Credibility distribution

Distribution of Tweet Sources' Credibility



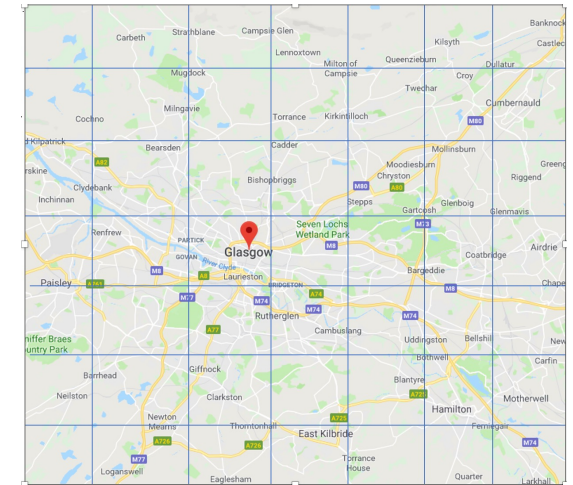
Weighted majority voting algorithm

- Vote from tweet t_i is weighted by

$$Vote(t_i^{l_i}, l_j) = \begin{cases} 1 & t_i^{l_i} = l_j \\ 0 & t_i^{l_i} \neq l_j \end{cases}$$

$$Location(t_{ng}) = \operatorname{argmax}_{l_j \in L} \left(\sum_{i=1}^N W_{t_i}(\alpha) * Vote(t_i^{l_i}, l_j) \right)$$

$$W_{t_i}(\alpha) = \alpha \cdot Credibility(s_i) + (1 - \alpha) \cdot Sim(t_i, t_{ng})$$





Part V

Evaluation methodology

Ideally

- We are aiming to assign location information to tweets
 - With no location information
- That is tweets with no coordinates field
 - **'geo_enabled': False, 'coordinates': null,**
- **Assign a location to the given tweet**

Our approach is

- **Develop a model, an approach, a technique ...**
 - IR based approach
- **Train and test them in a collection of known cases**
 - Ground truth
 - Gold standard
- **apply the technique to real-life scenarios and expect the technique/approach/method to work**
 - Similar to our evaluation case

Two scenarios

- **Tweets with geo-location**
- **Tweets without geo-location**
- **One problem though!!**
 - **We expect the characteristics of tweets in both cases/scenarios are the same**
 - **May not be true!!!**
 - Not all tweets are geo-tagged, which leads to sparser data sets
 - This affects the user geo-tagging process
 - **Still we proceed with our assumption and evaluate**
 - However, it is still an open problem to prove the assumptions are correct

Tweets without geo-location

- how could we verify this assignment is true or not?
- **Assume, we have**
 - 'place_coordinates': [[[-3.235601, 51.825903], [-3.235601, 53.22621], [-1.172136, 53.22621], [-1.172136, 51.825903]]],
 - Can we assign a fine-grained location?
- And if the assigned location within this box let us assume that it is correct
- Alternatively
 - Get author to say it is correct or not OR
 - pay people/workers to verify it (Crowdsourcing)

Let us look at geo-tagged tweets

- { '_id': '1409603292112502790', 'date': datetime.datetime(2021, 6, 28, 21, 3, 8), 'username': 'Natezero27', 'text': "Just posted a photo @ Fenn's, Whixall and Bettisfield Mosses National Nature Reserve [https://t.co/hkBcrDoPlj](\"https://t.co/hkBcrDoPlj\")", 'geo_enabled': True, 'coordinates': [-2.76446, 52.92165], 'location': None, 'place_name': 'West Midlands, England', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[-3.235601, 51.825903], [-3.235601, 53.22621], [-1.172136, 53.22621], [-1.172136, 51.825903]]], 'hashtags': [], 'mentions': [], 'source': 'Instagram' }

Data sets

- Ground truth of English geo-tagged tweets
- Chicago data
 - 131,273 tweets
 - March 2016
- New York city
 - 155,114 tweets (NY1)
 - March 2016
 - 1,318,065 tweets (NY2)
 - October 2014

Do we know the location?

- All tweets have a coordinates info
 - { '_id': '1409603292112502790', 'date': datetime.datetime(2021, 6, 28, 21, 3, 8), 'username': 'Natezero27', 'text': "Just posted a photo @ Fenn's, Whixall and Bettisfield Mosses National Nature Reserve [https://t.co/hkBcrDoPlj](\"https://t.co/hkBcrDoPlj\")", **'geoenabled': True, 'coordinates': [-2.76446, 52.92165]**, 'location': None, 'place_name': 'West Midlands, England', 'place_country': 'United Kingdom', 'country_code': 'GB', **'place_coordinates': [[[-3.235601, 51.825903], [-3.235601, 53.22621], [-1.172136, 53.22621], [-1.172136, 51.825903]]]**, 'hashtags': [], 'mentions': [], 'source': 'Instagram'}

We assume that

- Geotagged tweets & non-geo tagged tweets have same characteristics
 - Han, Cook, Baldwin 2014
- We also assume that we have city level information and focus on specific cities
 - New York City (USA)
 - Chicago (USA)

So...

- We know the exact location of each tweet
 - ... 'geoenabled': True, 'coordinates': [-2.76446, 52.92165], ...
 - An actual location. Of the tweet captured by the GPS
- During testing we ignore these information
 - In other words, we will not use this information in the prediction process
- That is, we assume this info do not exist and try to predict them
 - We have a predicted location
 - An actual location
- We can find the distance between these points

Data split

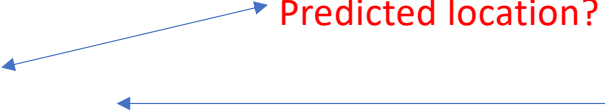
- First three weeks, training set
 - That means our models will not have any knowledge of validate or test cases.
- Last week data randomly grouped into
 - Validation &
 - Test set
- Validation is required to learn the credibility model

Data description

Dataset	Collection Time	Training	Validation	Testing
Chicago	March 2016	111,267	9,823	9,823
NYC_1	March 2016	128,746	13,184	13,184
NYC_2	October 2014	1,123,125	97,470	97,470

To make it clear...

We use this text

- ~~{ '_id': '1409603292112502790', 'date': datetime.datetime(2021, 6, 28, 21, 3, 8), 'username': 'Natezero27', 'text': "Just posted a photo @ Fenn's, Whixall and Bettisfield Mosses National Nature Reserve [https://t.co/hkBcrDoPlj](\"https://t.co/hkBcrDoPlj\")",~~
• 'geoenabled': True, 'coordinates': [-2.76446, 52.92165],
• 'location': None, 'place_name': 'West Midlands, England', 'place_country': 'United Kingdom', 'country_code': 'GB', 'place_coordinates': [[[~~3.235601, 51.825903~~], [~~3.235601, 53.22621~~], [~~1.172136, 53.22621~~], [~~1.172136, 51.825903~~]]], 'hashtags': [], 'mentions': [], 'source': 'Instagram'}'


Measures

- Average Error distance – smaller the better!
 - Predicted location and the actual coordinates of the tweet in our ground truth
 - Haversine formula is used!

Our aim is fine-grained localization!

- Accuracy@1km – higher the better
 - The accuracy of the model is measured by determining whether a predicted location lies within the radius of 1km from the real location –
 - how fine grained our prediction

We also need to know

- Recall/Coverage
 - Fraction of tweets in the test set that was geo-localised

Processing

- Square grid structure of 1 km
- Each grid has a document
 - Concatenated by all tweets from that area – baseline
 - Granular approach – we look at individual tweets
 - Grid containing '**coordinates**': [-2.76446, 52.92165],
 - "Just posted a photo @ Fenn's, Whixall and Bettisfield Mosses National Nature Reserve" + "Visiting beautiful Whixall and Bettisfield Mosses National Nature Reserve" +

- Indexing documents

"@ MusicNewsWeb <https://t.co/cGbkkwFHAS> Sheffield's Tramlines Festival 2021 will go ahead as a Government pilot <https://t.co/lonGcMLm74>"

- Usernames, hashtags were preserved as tokens
 - @ MusicNewsWeb
- Hyperlinks were removed
 - <https://t.co/lonGcMLm74> , <https://t.co/cGbkkwFHAS>
- Re-tweets preserved in the data set
 - We ignored it so far!
- Given a tweet
 - Retrieve most similar tweet in the collection
 - <http://lucene.apache.org/>

Results

Model		Av_error_km	Acc@1km %	Recall -%	
Baseline	TF-IDF	7.505	87.29	99.98	NYC_1
Voting Top 3	Alpha 0.0	2.491	71.83	58.65	
Voting Top 3	Alpha .75	3.179	66.19	71.05	
Model		Av_error_km	Acc@1km %	Recall %	
Baseline	TF-IDF	8.686	40.95	99.98	Chicago
Voting Top 3	Alpha 0.0	2.389	75.27	67.06	
Voting Top 3	Alpha .75	2.898	71.14	75.22	
Model		Av_error_km	Acc@1km	Recall	
Baseline	TF-IDF	13.981	19.54	84.59	NYC_2
Voting Top 3	Alpha 0.0	6.134	52.10	31.69	
Voting Top 3	Alpha .75	7.243	45.55	42.08	

Discussion of results

- Performance improved for small alpha values
 - Similarity between geo-tagged tweet has a big effect on model's accuracy and error distance
 - Amazing birthday party in night club at 7th avenue
 - Results in low similarity score
 - E.g., Happy birthday to my friend David
 - High similarity for
 - Birthday party at 7th avenue
- Chicago and NYC data sets shows similar performance differences
 - Portable across cities

Think of an alternative to credibility model?



Bias!!!!

In data
&
Other sources
Time ...

Bias in data

- **Bias in data** can be defined as a systematic distortion or imbalance in the representation of information
 - Data distribution across the grid
- that results in certain groups (**grid areas**) , outcomes, or perspectives being favored, underrepresented, or inaccurately portrayed.
- Bias can arise due to various factors, including how data is collected, pre-processed, or interpreted, and it can manifest in different forms depending on the context.
- **Selection Bias:**
 - When the sample data is not representative of the target area/region,
 - leading to skewed outcomes.
- Ideally, we expect the data to be distributed evenly across the grid
- If that is not true,
 - It might have affected the evaluation
 - It might affect the real-life usage

Distribution of tweets over the grids

- if all grids have enough tweets to formulate a language model that represents a grid,
- check for anomalies such as concentration of tweets or lack of tweets in certain grids and look for patterns in tweet distribution.
- Heatmap:
 - A heatmap analysis of the density of tweets in each grid location for the given dataset aims at investigating
- Histogram:
 - How are the tweets distributed across the grid?

London

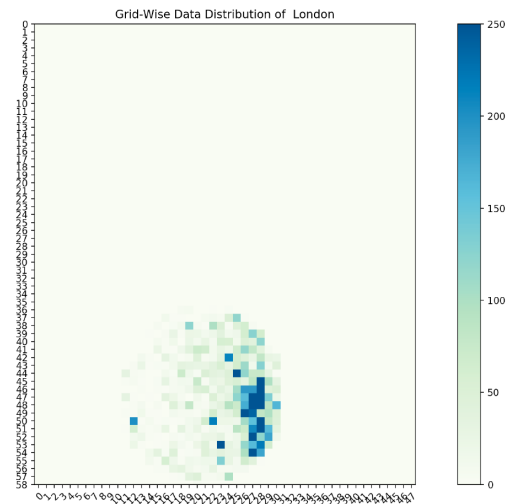


Fig 17: London- Heatmap of number of tweets per grid

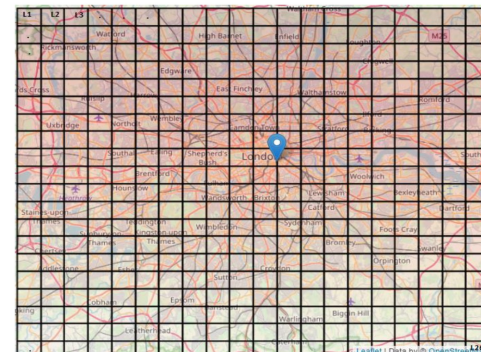


Fig 12: Diagrammatic representation of dividing London into 1Km2 Grids

How to form a heat map

- Count the number of tweets in each cell
- Then create a visualization of the grid
- Adapt the thresholds to show the changes

New York

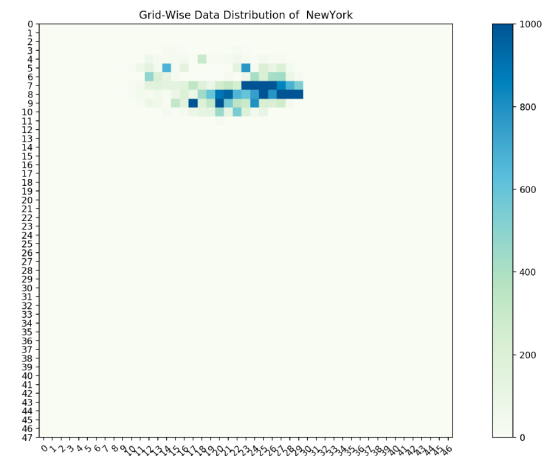
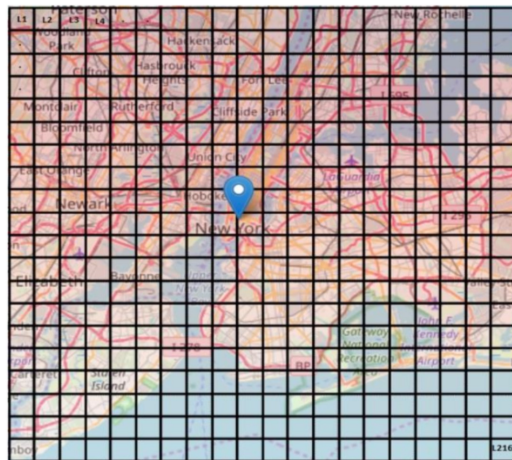


Fig 16: NewYork- Heatmap of number of tweets per grid

For each cell count the number of tweets

X-axis – the number of tweets

Y-axis – the number of cells/grids with that frequency

- Distribution of tweets – Histogram :
- A histogram with each bar showing the number of grids containing tweets in a given count range is used to
- analyse if the distribution of tweets is uniform and to investigate the range of variation of tweets in grids, etc.

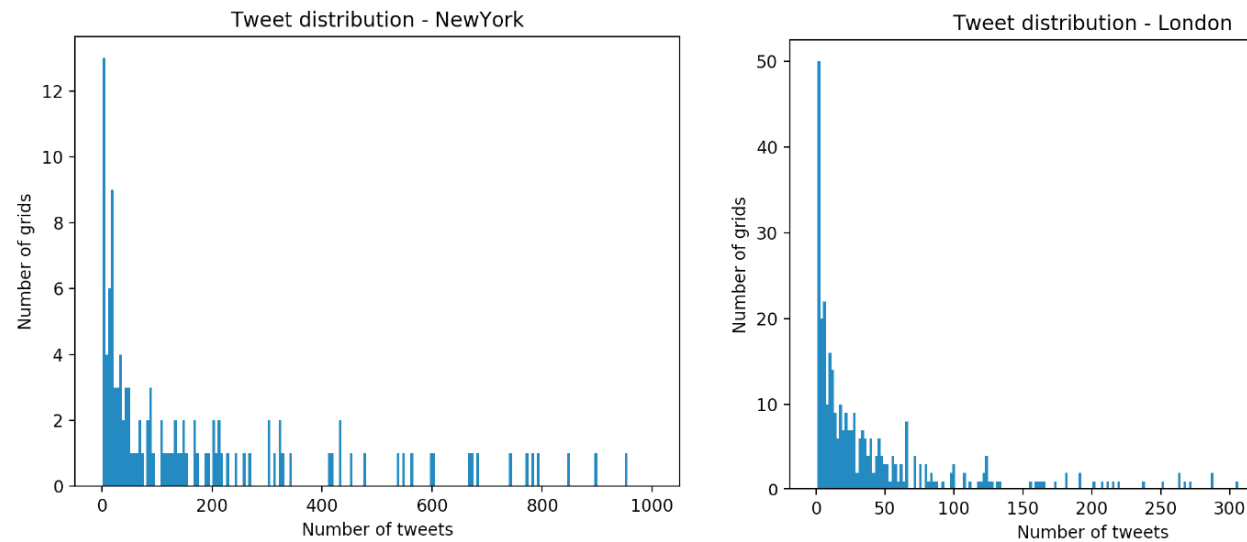


Fig 18: Tweet Distribution – NewYork and London

Analyzing couple of weeks data – Statistics!

Newyork -August	London-August
<i>Total Dataset size : 63000</i>	<i>Total Dataset size : 31913</i>
<i>Vocabulary Size : 42968</i>	<i>Vocabulary Size : 36194</i>
<i>Grid coverage : 128/2256</i>	<i>Grid coverage : 335/2832</i>
<i>Location Grids from 0 to 2255</i>	<i>Location Grids from 0 to 2831</i>
<i>Candidate grid count: 2256.000000</i>	<i>Candidate grid count: 2832.000000</i>
Tweets Distribution in Grid Statistics	Tweets Distribution in Grid Statistics
mean 27.925532	mean 11.268715
std 428.997121	std 254.116817
min 0.000000	min 0.000000
25% 0.000000	25% 0.000000
50% 0.000000	50% 0.000000
75% 0.000000	75% 0.000000
max 17912.000000	max 13225.000000

Table 14: Dataset and Tweet Distribution statistics -London and NewYork

Discussion - evaluation methodology

- Ground truths
 - Use all the geo-enabled tweets as ground truths
 - We know the location reasonably accurately
- Let us pretend that
 - we don't know the ground truth for portion of such ground truth data
 - Use that to compute measures
- Assume that geo-enabled and non-enabled tweets have same characteristics
- **Maybe we should use non-geo tagged tweets too**
 - How?

Summary

- Looked into geo-localisation
- Differentiated fine grained versus coarse grained differentiation
- Discussed a method to allocate fine grained localisation to tweets
- Discussed evaluation methodology